Ranga Rohit Nallamolu

n.rangarohit@gmail.com | Portfolio | linkedin | Github | +1 978 569 7460

SUMMARY

Data Engineer with 3+ years of experience building and optimizing batch and real-time pipelines across Databricks, Spark, AWS, Azure, and SAP Datasphere, driving enterprise forecasting and reporting for large-scale financial and operational data. Skilled in streaming architectures, cross-cloud integration, and data governance, with a track record of reducing reporting cycle times and improving reliability through scalable ETL frameworks. Extends core data engineering expertise into LLM-powered multi-agent systems for automated analytics.

EXPERIENCE

Data Engineer | *BigBlue InfoTech(Client:Mckesson), Dallas, TX* **Jan 2025 - Present**

- Architect multi-source ETL integration to drive the Enterprise Forecast Process Redesign (EFPR) — pulling financial and operational data from Databricks Lakehouse, Snowflake, and SAP BHP into a centralized SAP Datasphere layer, standardizing schemas and consolidation logic across systems to reduce end-to-end reporting cycle time by 25%.
- Support the Financial Development Plan (FDP) and EFPR initiatives by designing and automating scalable batch pipelines in Databricks (Spark) to process large-scale financial and application log datasets; orchestrate Databricks Jobs/Workflows to deliver forecast-ready Gold-layer datasets into SAP Datasphere for downstream ETL and SAC reporting.
- Improved pipeline efficiency and reduced compute costs by optimizing Spark job performance in Databricks (partition tuning, caching, cluster right-sizing), cutting processing time for large-scale financial datasets by 20%.
- Partnered with FP&A stakeholders to build KPI and forecast dashboards, using clean Databricks tables to power reports in SAP Analytics Cloud (SAC).
- Improved data accuracy by adding automated data quality checks in Databricks and reconciliation logic in SAP Datasphere, catching variances and keeping numbers consistent across systems.
- Implemented access controls and governance policies across Databricks and SAP Datasphere for data security and audit readiness.
- **Capabilities**: Databricks, SAP Analytics Cloud, SAP Datasphere, AWS, Apache Spark, SQL

Data Engineer | *University of Massachusetts, Lowell, MA* **Dec 2022 - May 2024**

- Architected scalable ETL pipelines using AWS Glue, Spark, and SQL to integrate multi-institutional data across Student, Finance, and Research systems into AWS S3 and Redshift, improving reliability, performance, and analytics readiness.
- Designed a scalable medallion architecture (Bronze/Silver/Gold layers) in S3 to support curated datasets for analytics, reducing redundant transformations and improving downstream report reliability.
- Built reusable data transformation frameworks in PySpark, enabling modular, testable ETL development and reducing new pipeline development time by 25%.
- Enforced data governance and security best practices through IAM policies, encryption (KMS), and role-based access control (RBAC) in Redshift to ensure FERPA compliance.
- Collaborated with Finance, IT, and Analytics teams to define KPIs and drive data-informed planning for enrollment, retention, and revenue.
- Developed Tableau dashboards to visualize key metrics and insights, contributing to a 17% increase in student retention.
- **Capabilities**: AWS, ETL, Python, SQL, Tableau, Spark

Research Assistant | *University of Massachusetts, Lowell, MA* **Dec 2022 - Nov 2023**

- Engineered data pipelines in Azure Data Lake and Azure Databricks to process and analyze 12 years of microbial shotgun metagenomics data, identifying a 70% correlation between Coprococcus bacteria and Type 2 diabetes risk.
- Developed machine learning models using Random Forest and XGBoost in Python to predict diabetes risk, achieving 85% accuracy and enhancing understanding of the gut microbiome's role in metabolic health.
- Analyzed microbial abundance, diversity indices, and butyrate production levels, uncovering the role of Coprococcus in maintaining insulin sensitivity and reducing Type 2 diabetes risk.
- Developed Power BI dashboards to visualize the impact of Coprococcus on diabetes, supporting data-driven decision-making in healthcare.
- Collaborated with research teams to consolidate findings, improving data representation and supporting diabetes prevention strategies.
- **Capabilities**: Azure Data Lake Storage, Azure Databricks, Python, R, Machine Learning, Statistical Analysis, Power BI

PROJECTS

AI-Driven Stock Trading & Research Platform *website Link*

- Built a real-time multi-agent trading platform with three coordinated AI layers — news scanning, recommendation generation, and trade execution — using a cost-tiered LLM pipeline (rule-based + Claude Haiku/Sonnet) to minimize API spend.
- Scans tickers for high-signal news (earnings, M&A, FDA, lawsuits) and generates BUY/SELL/HOLD recommendations with confidence scores and price targets by fusing fundamentals with sentiment..
- Runs a live agent debate every 10 minutes during market hours — bull/bear agents argue, a judge decides, a risk agent validates sizing, and an execution agent trades with automated stop-loss/take-profit monitoring.

Multi-Source Retail ETL Pipeline

- Designed a Bronze/Silver/Gold medallion pipeline in PySpark and Delta Lake to clean, deduplicate, and standardize multi-source retail transactional data (orders, customers, products).
- Orchestrated the pipeline using Databricks Workflows with Unity Catalog governance, producing aggregated Gold tables for revenue by region, customer lifetime value, and top products.
- Improved data reliability and reduced redundant transformations, enabling consistent, trustworthy reporting through Power BI dashboards.

SKILLS

Programming Languages: Python, Scala, Java, R

Cloud & Platforms: AWS, Azure, Databricks, Snowflake, SAP Datasphere, SAP Analytics Cloud

Data Engineering: ETL/ELT, Medallion Architecture, Data Governance, DBT

AI/ML: LLM Multi-Agent Systems, Claude API, Machine Learning

BI & Databases: Power BI, Tableau, PostgreSQL, MySQL, Redshift

EDUCATION

University of Massachusetts Lowell, MA

Master of Science in Computer Science **Sep 2022 - May 2024**

Christ University Bangalore, India

Bachelor of Technology in Computer Science **Aug 2017 - May 2021**